

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2014

CL402-Computer Aided Structural Analysis

Date: 21.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) For the iso-parametric quadrilateral elements shown in **figure: 1**, determine the local co-ordinates of the point Q which has Cartesian co-ordinates (7, 4). [07]

Q - 2 Attempt any two: [14]

- (a) Analyze the structure using Flexibility method shown in **Figure: 2**. All members are assumed to have the same axial rigidity EA. Determine B_{MS} , B_{RS} matrices.
- (b) Analyze the two span continuous beam using flexibility method member approach shown in **figure: 3**. Derive the B_{MS} and B_{RS} matrix for the beam.
- (c) Analyze the frame shown in **figure: 4**. Find B_{MS} and B_{RS} matrices using flexibility method member approach.

Q - 3 (a) Assemble the Jacobian Matrix corresponding to (0.57735, 0.57735) for element shown in **figure: 5**. [07]

(b) Using generalized co-ordinate approach, find the shape functions for two-noded bar element shown in **figure: 6**. [07]

OR

(c) Derive the Shape Function for a one Dimensional body as shown in **figure: 7** using Lagrange functions. Draw the variations of shape function. [07]

SECTION - II

Q - 4 (a) Analyze the beam using stiffness method system approach. Refer **figure: 8**. [07]

Q - 5 (a) A continuous beam ABCD (Refer **Figure: 9**) is carrying distributed load of 5 kN/m as shown in figure 9. Compute reactions due to following support settlements. Take EI as constant. Support B - 0.005m vertically downward and Support C - 0.010m vertically downward [14]

OR

Q - 5 (a) Analyse the beam as shown in **figure: 10**. Take EI as constant. [14]

Q - 6 (a) Analyse the truss as shown in **figure: 11** if, temperature of AB and AC is increased [14]

by 50°C and 25°C respectively. Take $AE = 32000\text{ kN}$; $\alpha = 0.00002$. Length of BC 3 mm more.

OR

Q - 6 (a) Find the support reaction for the frame shown in Figure: 12. by stiffness method. [06]

Take $A = 0.08\text{ m}^2$ and $I = 1.067 \times 10^{-3}$.

(b) Write the member stiffness matrix in member axes and rotation matrix for the space [08]
frame member.

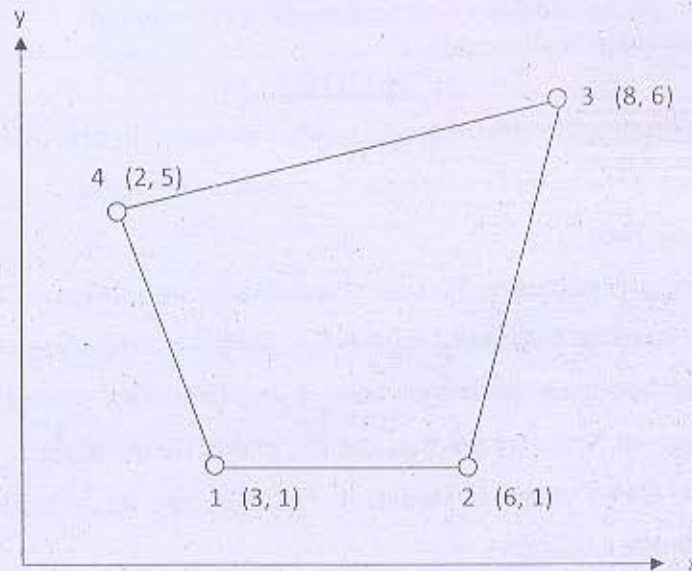


Figure: 1

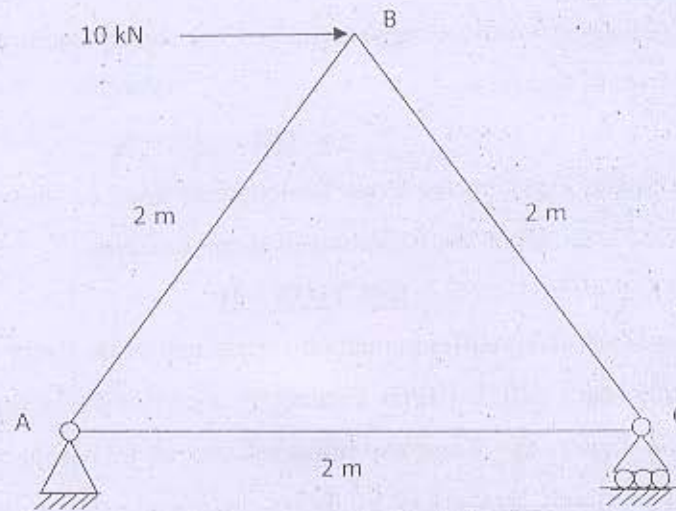


Figure: 2

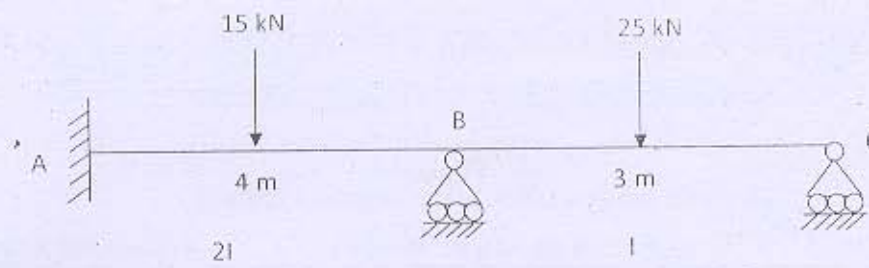


Figure: 3

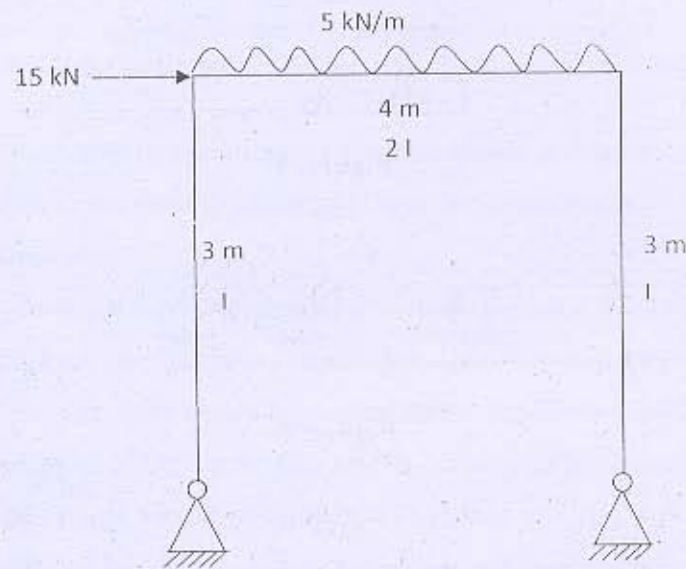


Figure: 4

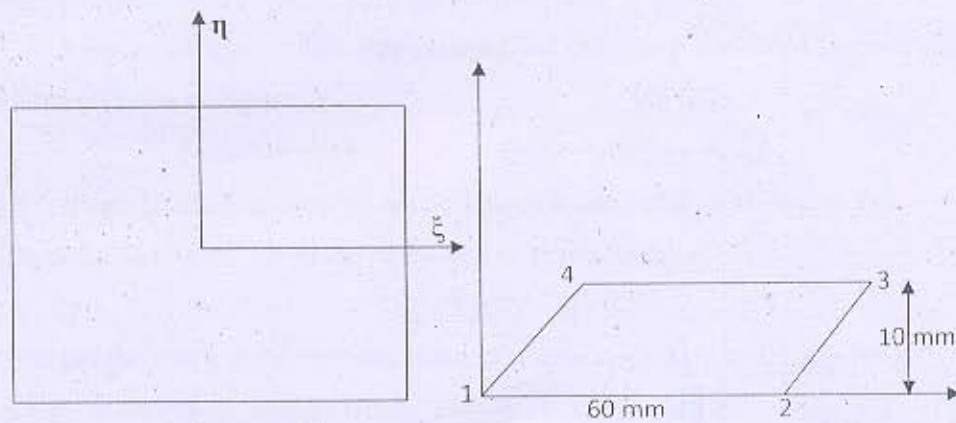


Figure: 5



Figure: 6

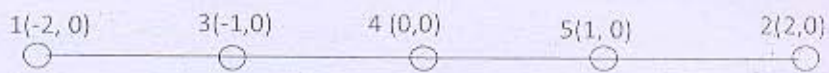


Figure: 7

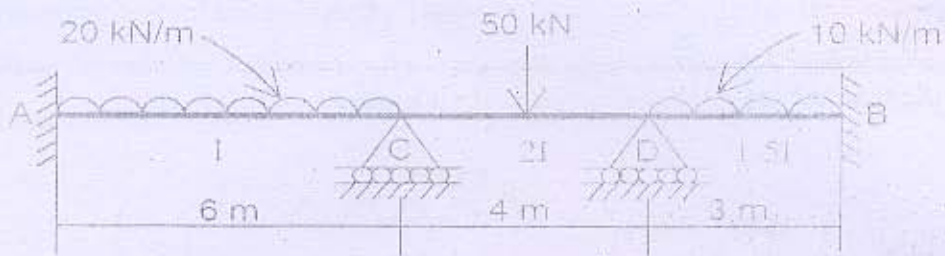


Figure: 8

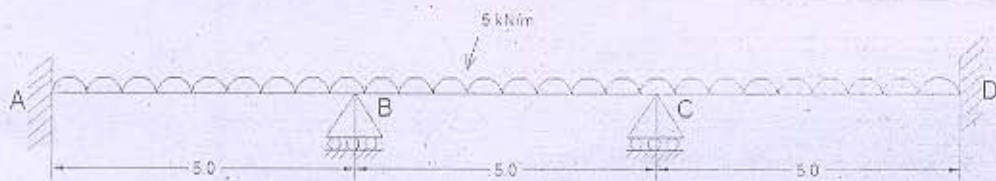


Figure: 9



Figure: 10

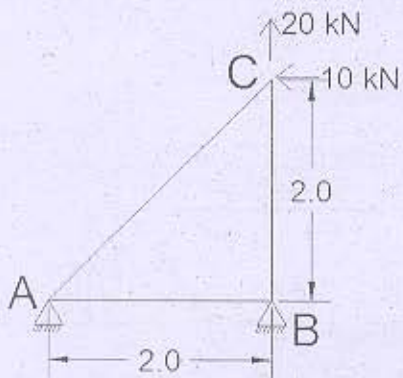


Figure: 11

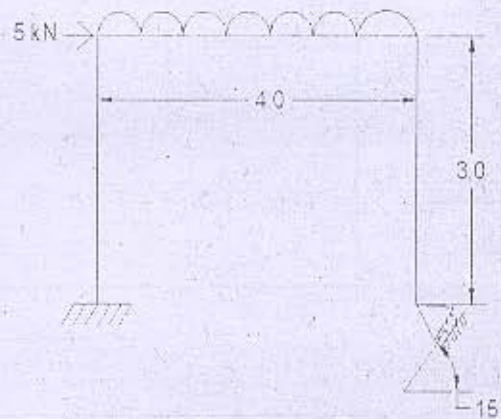


Figure: 12
